

5. Objectives

- Explain how to clean up and push back to remote Git

In this hands-on lab, you will learn how to:

- Execute steps involving clean up and push back to remote Git.

Prerequisites

The following are the pre-requisites to complete this hands-on lab:

- Hands-on ID: **“Git-T03-HOL_002”**

Notes*:

Please follow the below steps for creating a free account in GitHub.

Do not use cognizant credentials to login to GitHub.

Estimated time to complete this lab: **10 minutes.**

Please follow the instructions to complete the hands-on. Each instruction expects a command for the Git Bash.

1. Verify if master is in clean state.
2. List out all the available branches.
3. Pull the remote git repository to the master
4. Push the changes, which are pending from **“Git-T03-HOL_002”** to the remote repository.
5. Observe if the changes are reflected in the remote repository.

Answer:

How to clean up and push back to remote Git

1. Verify clean working state

- Run git status to check if there are any uncommitted changes.
- If there are changes, either commit them using git add . and git commit -m "message", or discard them.
- This ensures your local branch is “clean” before pushing.

2. Switch to the correct branch

- Use `git checkout <branch-name>` to move to the branch you want to update (e.g., main or your feature branch).

3. Pull latest changes from remote

- Run `git pull origin <branch-name>` to sync your local branch with the latest version from the remote repository.
- This helps avoid conflicts when pushing.

4. Push your branch to remote

- If your branch has new commits, run:
`git push origin <branch-name>`
- This sends your local commits to the remote repository (GitHub, GitLab, etc.).

5. Verify on remote

- Open your repository in GitHub/GitLab.
- Check that your commits and files are visible on the correct branch

```
anjub@Meghumi MINGW64 ~
$ cd 6410978

anjub@Meghumi MINGW64 ~/6410978 (Git-T03-HOL_002)
$ git branch -a
* Git-T03-HOL_002
  main
  remotes/origin/Git-T03-HOL_002
  remotes/origin/HEAD -> origin/main
  remotes/origin/main

anjub@Meghumi MINGW64 ~/6410978 (Git-T03-HOL_002)
$ git status
On branch Git-T03-HOL_002
nothing to commit, working tree clean

anjub@Meghumi MINGW64 ~/6410978 (Git-T03-HOL_002)
$ git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)

anjub@Meghumi MINGW64 ~/6410978 (main)
$ git status
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)

nothing to commit, working tree clean

anjub@Meghumi MINGW64 ~/6410978 (main)
$ git add .

anjub@Meghumi MINGW64 ~/6410978 (main)
$ git commit -m "Clean up main branch"
On branch main
Your branch is ahead of 'origin/main' by 1 commit.
(use "git push" to publish your local commits)

nothing to commit, working tree clean

anjub@Meghumi MINGW64 ~/6410978 (main)
$ git branch -a
* Git-T03-HOL_002
  main
  remotes/origin/Git-T03-HOL_002
  remotes/origin/HEAD -> origin/main
  remotes/origin/main
```

```
anju@Meghumi MINGW64 ~/6410978 (main)
$ git branch -a
Git-T03-HOL_002
* main
  remotes/origin/Git-T03-HOL_002
  remotes/origin/HEAD -> origin/main
  remotes/origin/main

anju@Meghumi MINGW64 ~/6410978 (main)
$ git pull origin main
From https://github.com/anjanaswaminathan/6410978
* branch      main       -> FETCH_HEAD
Already up to date.

anju@Meghumi MINGW64 ~/6410978 (main)
$ git checkout Git-T03-HOL_002
Switched to branch 'Git-T03-HOL_002'

anju@Meghumi MINGW64 ~/6410978 (Git-T03-HOL_002)
$ git push origin Git-T03-HOL_002
Everything up-to-date

anju@Meghumi MINGW64 ~/6410978 (Git-T03-HOL_002)
$ git checkout main
Switched to branch 'main'
Your branch is ahead of 'origin/main' by 1 commit.
(use 'git push' to publish your local commits)

anju@Meghumi MINGW64 ~/6410978 (main)
$ git merge Git-T03-HOL_002
Already up to date.

anju@Meghumi MINGW64 ~/6410978 (main)
$ git push origin main
Total 0 (delta 0); reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/anjanaswaminathan/6410978.git
3b55e4c..18e417e  main -> main
```